Episode 1: The Route Map

6 questions, in the order the audio asks them. When it says "Pause the audio now," pause, answer, then play.

STEP ZERO. THE HIGH-PILED GATE.

1. SAY IT · NO BOOK A Class Three commodity, stored ten feet high. Under the 2022 edition, is it high-piled or low-piled, and where do you go next?

STEP TWO. CHAPTER TWENTY, GENERAL REQUIREMENTS FOR STORAGE.

2. FIND IT · BOOK OPEN A customer adds a building of baled cotton. Which section in Chapter Twenty covers baled cotton, and where does it send you for protection?

3. SAY IT · NO BOOK Name the three things you write down before you leave Chapter Twenty.

STEP THREE. SCREEN THE BUILDING.

4. SAY IT · NO BOOK The roof slopes three inches in twelve. Which sprinkler technology does that take off the table, and which rule says so?

STEP FOUR. PICK THE PROTECTION CHAPTER.

5. SAY IT · NO BOOK You've picked your protection chapter, and its table doesn't cover your commodity, arrangement, and height. What are your two moves?

STEP FIVE. HOSE ALLOWANCE AND DURATION.

6. FIND IT · BOOK OPEN Your warehouse is ESFR. Where do you find the hose stream allowance and the water supply duration?

FSM STORAGE PROCEDURE · NFPA 13 (2022) · QUESTIONS ONLY

Episode 2: Classify the Commodity

9 questions, in the order the audio asks them. When it says "Pause the audio now," pause, answer, then play.

WARM-UP

1. SAY IT · NO BOOK Group A plastic, stored eight feet high. Under the 2022 edition, is that high-piled storage?
2. SAY IT · NO BOOK You need the hose stream allowance and the water supply duration for a storage design. Which table?
3. SAY IT · NO BOOK Rubber tires in storage. Do their design criteria come from Chapter Twenty-Six?

STEP A. FIND THE PLASTIC.

4. FIND IT · BOOK OPEN A pallet load of products made of silicone rubber. What is it protected as?

STEP B. THE CLASS ONE THROUGH FOUR LADDER.

5. SAY IT · NO BOOK A cartoned product that's ten percent nonexpanded Group A plastic by weight. What class is it?

STEP C. GROUP A: CARTONED OR EXPOSED, EXPANDED OR NONEXPANDED.

6. FIND IT · BOOK OPEN An exposed product, no carton. Its only plastic is expanded Group A, at twenty percent by volume. Is it Class Four, or something else?

STEP D. CHECK THE PALLET.

7. FIND IT · BOOK OPEN A cartoned Class Three product on unreinforced H-D-P-E pallets, permanently marked. The building has ceiling-only spray sprinklers, K sixteen point eight. Does the commodity class go up?

STEP E. CHECK FOR ENCAPSULATION.

8. SAY IT · NO BOOK Palletized storage, encapsulated, Class Three, eighteen feet high. What's the minimum K-factor, and what's the density?

STEP F. CHECK FOR MIXED COMMODITIES.

9. FIND IT · BOOK OPEN A building whose ceiling protection is designed for Class Two. The owner wants to keep some Group A pallet loads scattered through a thirty-thousand-square-foot area. How many loads are allowed?

Episode 3: Arrangement and Rack Type

9 questions, in the order the audio asks them. When it says "Pause the audio now," pause, answer, then play.

WARM-UP

1. SAY IT · NO BOOK Group B plastic, and free-flowing Group A plastic. What are they protected the same as?

2. SAY IT · NO BOOK An exposed product, Group A plastic, with more than five, up to twenty-five percent by volume of expanded plastic. What is it?

3. SAY IT · NO BOOK A Class Two product on polypropylene pallets with no marking, under K eleven point two ceiling sprinklers. What class do you protect it as?

STEP B. NAME THE RACK TYPE.

4. SAY IT · NO BOOK Two double-row racks, each eight feet deep, with a three-foot aisle between them. What rack type?

5. SAY IT · NO BOOK A single-row rack is set eighteen inches from a wall. What rack type is it treated as, and where is its longitudinal flue?

STEP C. CHECK THE FLUES.

6. FIND IT · BOOK OPEN A double-row open rack holds a Class Three commodity, stored to twenty-two feet. Is a longitudinal flue required?

STEP D. OPEN RACK OR SOLID SHELF.

7. FIND IT · BOOK OPEN A double-row rack has no shelves. Storage is twenty-four feet. The operator pushed the loads back to back, so there's no longitudinal flue. Transverse flues are four feet apart. Open rack or solid-shelf rack?

8. FIND IT · BOOK OPEN A double-row rack has one solid steel shelf, fifteen square feet in area. The rest of the rack is open, with its six-inch longitudinal flue and transverse flues in place. Does that shelf make it a solid-shelf rack?

STEP E. THE HEIGHTS.

9. SAY IT · NO BOOK Most of the year a building stores to twenty-two feet. Every peak season, the same racks are stacked to twenty-eight feet. What storage height do you design for?

Episode 4: CMDA

9 questions, in the order the audio asks them. When it says "Pause the audio now," pause, answer, then play.

WARM-UP

1. SAY IT · NO BOOK Two ten-foot double-row racks stand with a three-foot aisle between them. What rack type does the code see?

2. SAY IT · NO BOOK Double-row open racks. Above what storage height is a six-inch longitudinal flue required?

3. SAY IT · NO BOOK A thirty-foot double-row rack with no shelves in it at all. The owner pushes the loads back to back and closes the longitudinal flue. Open rack, or solid shelf?

STEP A. FIND YOUR SECTION.

4. SAY IT · NO BOOK Class Three commodity in double-row racks, thirty feet of storage, CMDA. Where does Chapter Twenty-One send you, and what comes with it?

STEP B. READ THE TABLE.

5. FIND IT · BOOK OPEN Double-row racks, Class Two, not encapsulated, twenty-four feet of storage, eight-foot aisles, ordinary-temperature sprinklers. What density does the table give?

6. SAY IT · NO BOOK Twenty-four feet of storage, not encapsulated. Class Four in double-row racks, and Class Two in multiple-row racks sixteen feet deep with eight-foot aisles. Which one can CMDA protect with ceiling sprinklers only?

STEP C. THE ADJUSTMENTS.

7. SAY IT · NO BOOK The encapsulation adder. What rack type and height band does it cover, and what does it add for Class One through Three, and for Class Four?

STEP D. RUN OUR WAREHOUSE.

8. FIND IT · BOOK OPEN Cartoned Group A plastic, thirty feet of storage, and the designer puts the top in-rack level at twenty-two feet. What ceiling design do you use?

STEP E. HOSE AND DURATION.

9. FIND IT · BOOK OPEN A different job, CMDA, with a ceiling design area of fourteen hundred square feet. What hose stream and duration?

FSM STORAGE PROCEDURE · NFPA 13 (2022) · QUESTIONS ONLY

Episode 5: CMSA and ESFR

9 questions, in the order the audio asks them. When it says "Pause the audio now," pause, answer, then play.

WARM-UP

1. SAY IT · NO BOOK Our warehouse under CMDA. Cartoned Group A plastic, thirty feet of storage. Can the ceiling sprinklers do it alone?
2. SAY IT · NO BOOK Same warehouse, CMDA with in-racks. The top in-rack level sits at twenty-five feet, so there are five feet of storage above it. What's the ceiling design?
3. SAY IT · NO BOOK That CMDA ceiling design covers two thousand square feet. What's the hose allowance and duration?

STEP A. SCREEN THE BUILDING.

4. SAY IT · NO BOOK An ESFR area sits right next to an area of standard-response sprinklers. What has to go between them, and how big?

STEP B. CMSA.

5. FIND IT · BOOK OPEN Our warehouse. Cartoned nonexpanded Group A plastic, double-row racks, thirty feet of storage, thirty-five-foot ceiling, CMSA. What head, how many design sprinklers, and what pressure?

STEP C. ESFR.

6. FIND IT · BOOK OPEN Our warehouse. Thirty feet of storage under a thirty-five-foot ceiling, ESFR ceiling-only, K twenty-five point two pendent. What's the minimum pressure?
7. FIND IT · BOOK OPEN Same ceiling, thirty-five feet. Now the commodity is cartoned expanded plastic. What does ESFR ceiling-only give you?

STEP D. ESFR WITH ONE LEVEL OF IN-RACKS.

8. SAY IT · NO BOOK Foam cups again, cartoned expanded plastic, thirty-five-foot ceiling. The ceiling-only table is all dashes. Can you add one level of in-racks and use the ESFR plus in-rack table?

STEP E. HOSE AND DURATION.

9. SAY IT · NO BOOK ESFR with twelve design heads. What's the hose allowance and duration? And standard-spacing CMSA with fifteen?

FSM STORAGE PROCEDURE · NFPA 13 (2022) · QUESTIONS ONLY

Episode 6: In-Racks, Alternatives, Special Designs

9 questions, in the order the audio asks them. When it says "Pause the audio now," pause, answer, then play.

WARM-UP

1. SAY IT · NO BOOK Table twenty-three point three point one, the ESFR ceiling-only table. What is it keyed on?

2. SAY IT · NO BOOK In an ESFR ceiling design, how many sprinklers are in the design area, and how are they laid out?

3. SAY IT · NO BOOK CMSA with fifteen design heads. What hose stream and what duration?

STEP B. PICK THE ARRANGEMENT.

4. FIND IT · BOOK OPEN Same cartoned Group A plastic over twenty-five feet, but you pick option two instead of option one. How far apart can the in-rack levels be, vertically?

5. SAY IT · NO BOOK Our warehouse, option one, in-rack levels at ten and twenty feet only, thirty feet of storage. What's the ceiling density?

STEP D. IN-RACKS UNDER ESFR.

6. FIND IT · BOOK OPEN ESFR at the ceiling, open racks, one level of in-racks. The drawing shows K five point six standard-response in-racks. What has to change?

7. FIND IT · BOOK OPEN Our warehouse: cartoned nonexpanded Group A, thirty feet of storage, thirty-five-foot ceiling, ESFR with one level of in-racks. The ceiling heads are K sixteen point eight pendent. How many ceiling heads, at what pressure?

STEP E. THE HYDRAULICS.

8. SAY IT · NO BOOK The in-rack demand and the ceiling demand meet at the connection. What happens to the flows, and what happens to the pressures?

9. FIND IT · BOOK OPEN You're using section twenty-five point six, in-racks independent of the ceiling. Cartoned nonexpanded Group A plastic, option two B. What's the maximum vertical spacing between in-rack levels?